Wireless InSite®

Culvert Modeling Application Guide

May 2012

Contents

1. Culvert Modeling Application Overview **1**

1.1 Introduction to Culvert Modeling Application…… 1

2. Geometry Set-Up **2**

2.1 Creating a Culvert……... 2

2.2 Culvert Geometry Window…… 3

2.2.1 Declaring the Origin…… 4

2.2.2 Rotating the Culvert……. 4

2.2.3 Setting Dimensions…….. 5

2.2.4 Bending the Culvert……. 6

2.2.5 Materials..…… 8

2.2.6 Enclosing Box…….. 9

3. Study Area Set-Up **10**

3.1 Study Area Geometry…… 10

3.2 Prop. Models; Ray Spacing and Suggested Interaction……... 11

3.2.1 Table of Minimal Suggested Interactions……..... 11

4. Transmitter and Receiver Set-Up **12**

Appendix

A. Propagation Model Analysis **13**

A.1 Comparing X3D and Full3D with XFdtd®……...…… 13

1. Culvert Modeling Application Overview

1.1 Introduction to Culvert Modeling Application

Culvert Modeling is an application, within Wireless InSite, that was created to optimize the accuracy of solving electric fields inside hollow structures. The specific propagation models which are compatible with Remcom's Culvert Modeling Application are Full3D and X3D. This application guide assumes that the user is familiar with Remcom's Wireless InSite. If you are unfamiliar with common terms throughout this guide, it is suggested that you review the Wireless InSite® Reference Manual. This application guide was constructed to explain the extended Culvert Modeling options in an example format. *Appendix A. Propagation Model Analysis* will explain the validation study of Remcom's Wireless InSite® propagation models Full3D and X3D and Remcom's full wave solver XFdtd® Release 7 (XF7). Many of the option parameter recommendations discussed in this guide were developed from this study.

For the Culvert Modeling application, a culvert is defined as a tunnel-like structure with an opening at both ends of the tunnel. The culvert may only possess 1 of the following bends/turns: 90 degree, 45 degree, or 0 degree (no turns).

2. Geometry Set-Up

2.1 Creating a Culvert

Within the Wireless InSite Graphical User Interface (GUI), a culvert is considered to be a "Feature". There are several options that the user may specify to customize their culvert. These customizations will have an impact on the run-time and results. Before beginning your first Culvert Modeling scenario, Remcom's engineers recommend setting up, simulating, and briefly analyzing, a few different culvert geometries. This will establish a speed/accuracy comfort level for the user to apply toward future projects. The specific impacts that these options will have on run-time and results are minimal and will be discussed in future chapters.

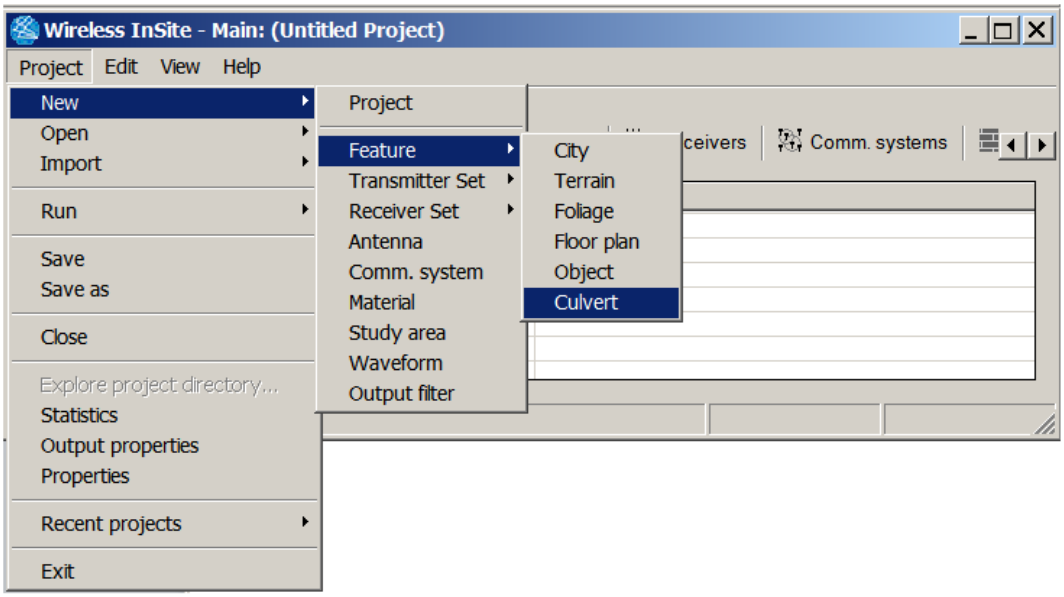


Figure 1.1 – Selecting a Culvert Feature

To create a new Culvert feature, from the Main window:

1. Click on *Project* → *New* → *Feature* → *Culvert* (see Fig. 1.1 above)
2. The Culvert Geometry window will open up automatically. Specify your culvert geometry in the Culvert Geometry window as in Fig 1.2, below. Please see the following section for specific information about the Culvert Geometry window.

   Note: Do not click on 'OK' until all of your geometry has been specified. You cannot access the Culvert Geometry window to make changes after clicking 'OK'.

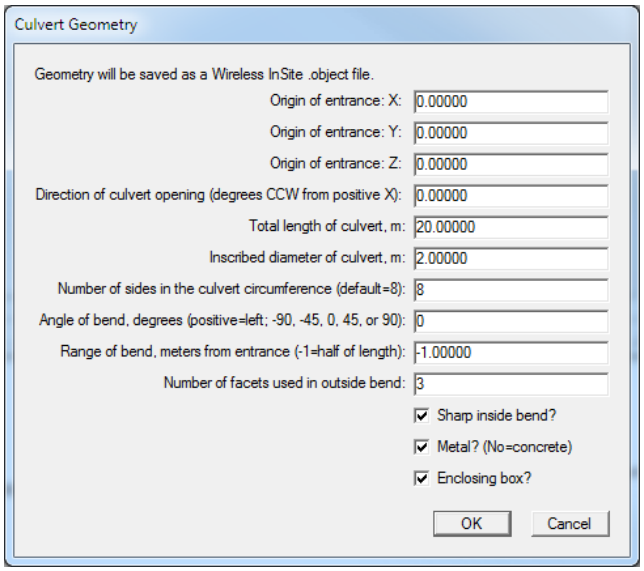


Figure 1.2 – Culvert Geometry window

3. Click OK.

2.2 Culvert Geometry Window

The Culvert Geometry window allows the user to control many aspects of their culvert, such as: where it is located, object dimensions, how it is constructed, the type of angles/bends, the material, and much more. This section will explain these options and offer some guidance to set up the culverts. **Chapter 3: Study Area Set-Up** will offer more suggestions to help obtain optimum results.

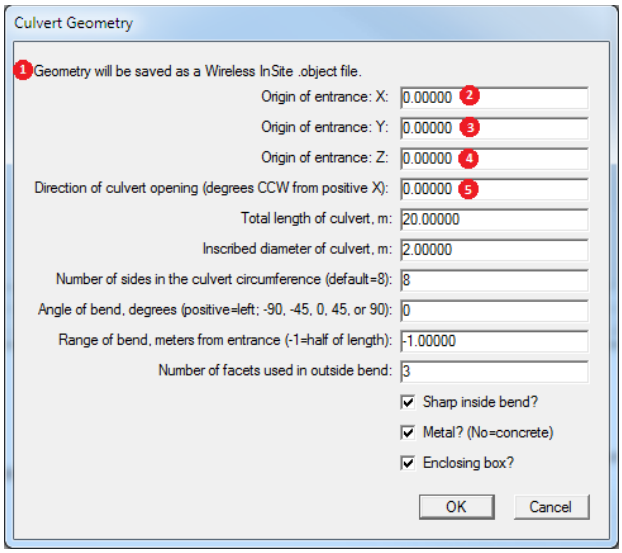


Figure 2.2.1 – Placing the culvert in the scene

The user will determine where the culvert is placed by selecting the origin and a "Direction of culvert opening". Once the Culvert's geometry is set up it will be saved as a Wireless InSite .object file in the project's directory. This is displayed as 1 in the Culvert Geometry window in Figure 2.2.1, above. The culvert's origin will always be at the center of the culvert's opening entrance. The floating red dot represents the culvert's origin, in the figure below.

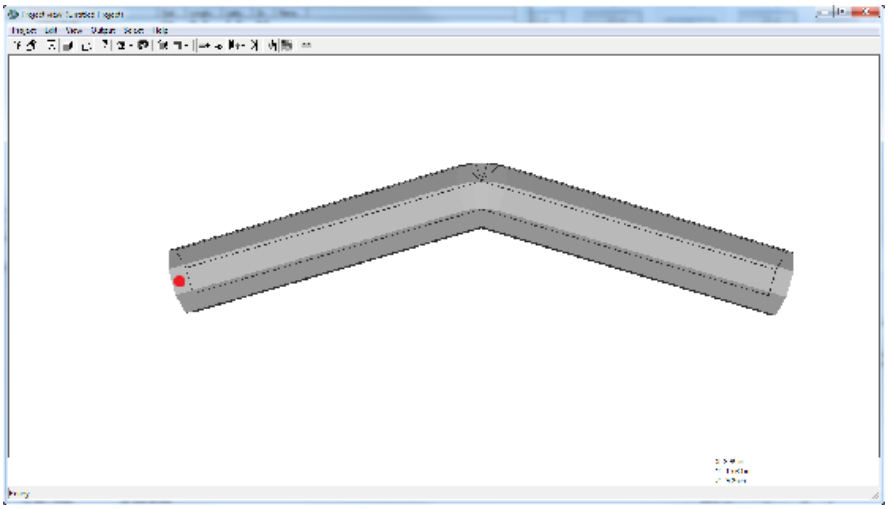


Figure 2.2.2 - Shows origin at center of opening

2.2.1 Declaring the Origin

The user will select the origin by entering the proper coordinates in the Culvert Geometry window labeled *Origin of entrance: X:*, 2, *Origin of entrance: Y:*, 3, and *Origin of entrance: Z:*, 4.

2.2.2 Rotating the Culvert

The user may also rotate the culvert in the Culvert Geometry window by entering an angle (in degrees) in the box labeled: "*Direction of culvert opening (degrees CCW from positive X)*", 5. Considering a bird's eye view of the culvert (looking toward the –z axis), the rotation angle entered in the box will follow the same orientation as the culvert in Fig. 2.2.3, below.

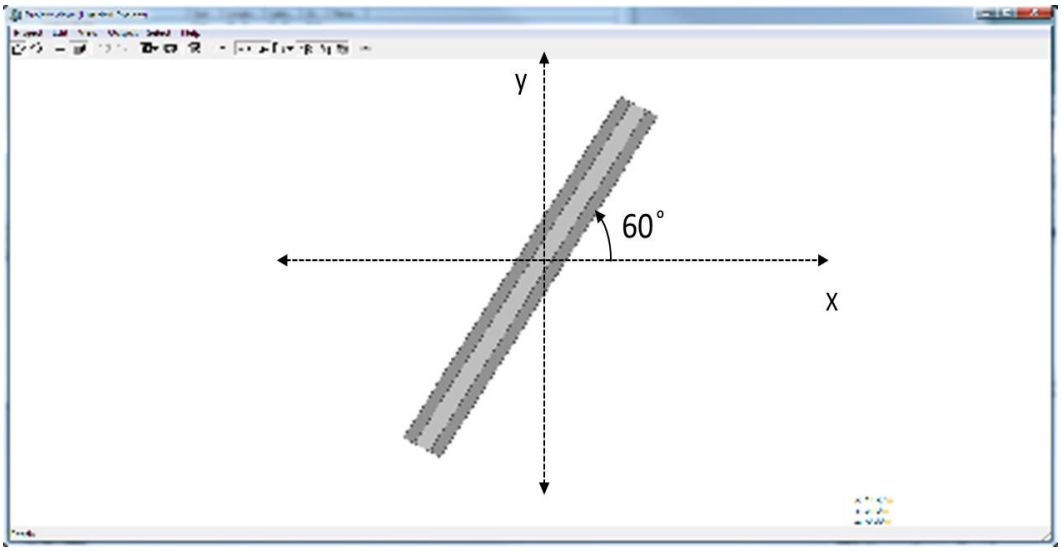


Figure 2.2.3 – A culvert rotated 60 degrees.

You may also rotate the culvert in the Wireless InSite Main window, as you would any Wireless InSite feature. Please see the reference manual for rotation instructions using the Main window.

2.2.3 Setting Dimensions

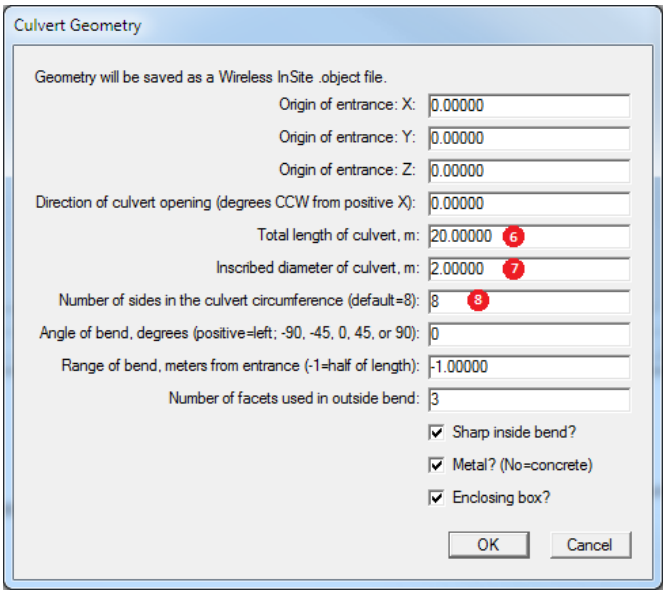


Figure 2.2.4 – Setting culvert dimensions

The box following *Total length of Culvert, m:*, 6, requires the length of the culvert, measured from the origin and extending to the tail opening of the culvert on the opposite side. This value should be in meters.

Inscribed diameter of culvert, m , 7, represents the diameter, or width, of the culvert's mouth. This value is also in meters.

The option *Number of sides in the culvert circumference (default=8):*, 8, represents the number of segments used to define the culvert's cross section. Remcom recommends no fewer than 4, and no more than 12 sides.

2.2.4 **Bending the Culvert**

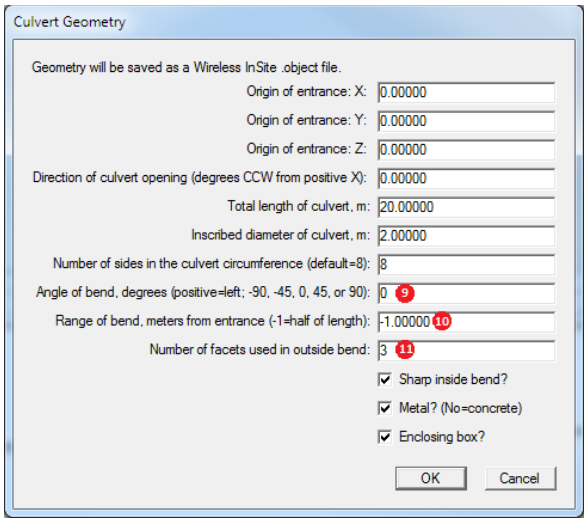


Figure 2.2.5 – Bending Culverts

Culverts may be created with a single bend. The angle of the bend must be 0, 45, or 90 degrees. Negative angles bend to the right. 0 represents a straight culvert. These bends are created using the box labeled, *Angle of bend, degrees (positive=left: -90, -45, 0, 45, or 90)*, 9.

The user may place the bend at any location of the culvert. This is done by entering a length, in meters, into the box labeled, *Range of bend, meters from entrance (-1=half of length):*, 10. To place the bend at the midpoint of the culvert, enter -1.

The final user defined box, 11, is used to create the outer edge of the bend. The value entered in the box labeled, *Number of facets used in outside bend:*, will allow the user to create a more rounded outside bend. More facets will result in a more rounded bend. A zero results in a picture-frame style bend. Below are a few examples of bends that could be created. Notice the difference in facets used at the bends. Bends with multiple facets may require the user to designate more diffractions in the study area.

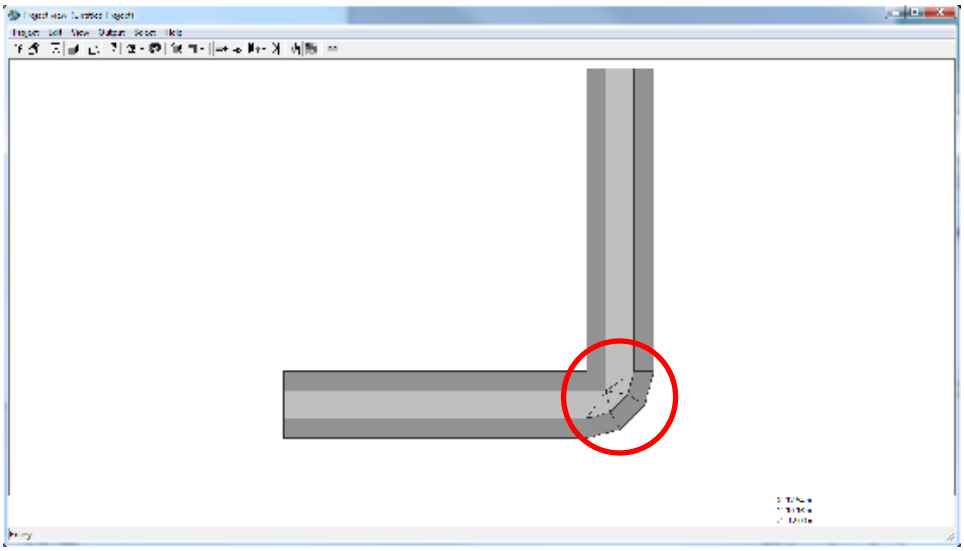


Figure 2.2.6 - 90 deg. Bend, with 3 facets on the outside edge of bend.

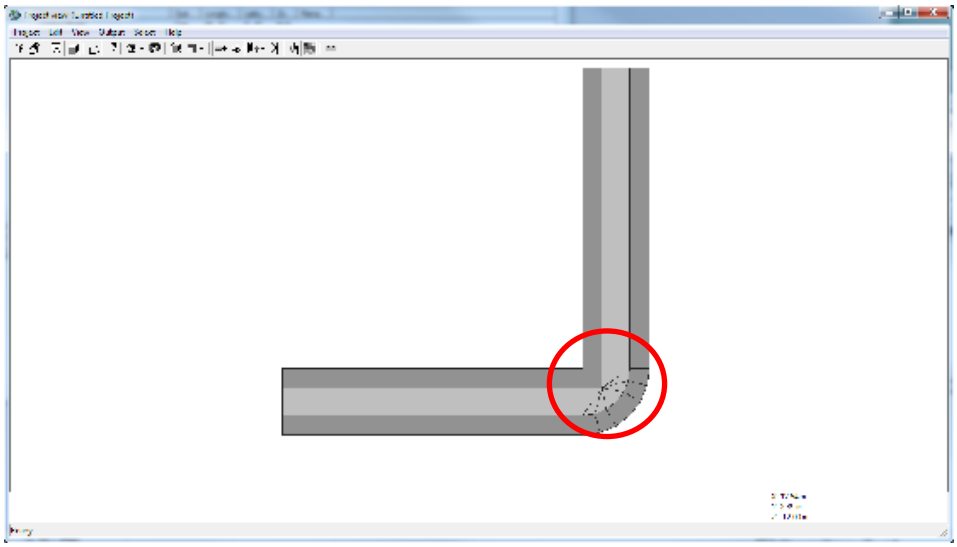


Figure 2.2.7 - 90 deg. Bend, with 6 facets on the outside edge of bend.

It is also possible for the user to control the bend of the inside. To make the bend sharp, click the box beside *Sharp inside bend?*. If this box is not checked, Remcom's Culvert Modeling tool will create the inside bend with as many facets as there are on the outside bend. An example of a sharp inside bend and a gradual inside bend are shown in the two pictures below.

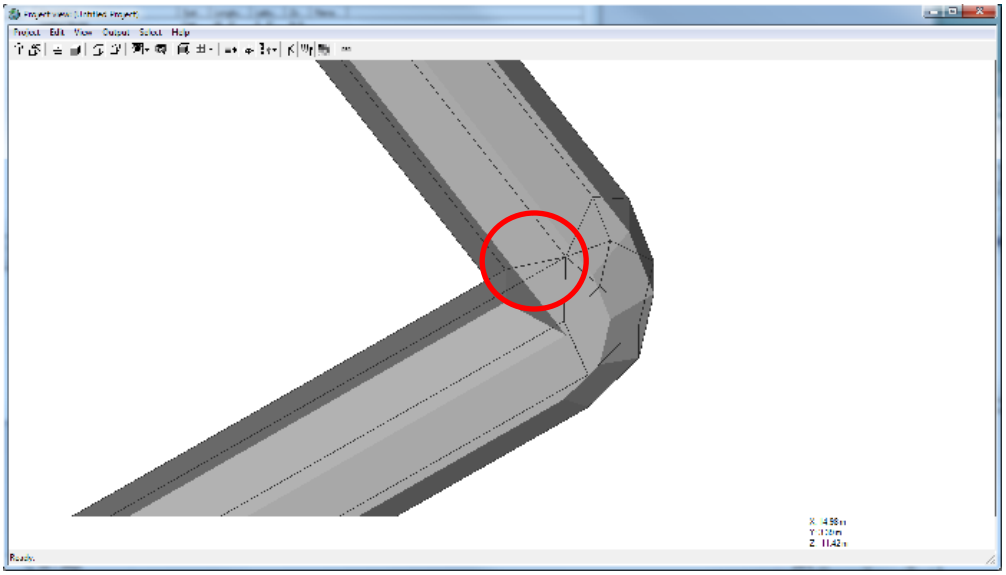


Figure 2.2.8 – *Sharp inside bend?* - Unchecked

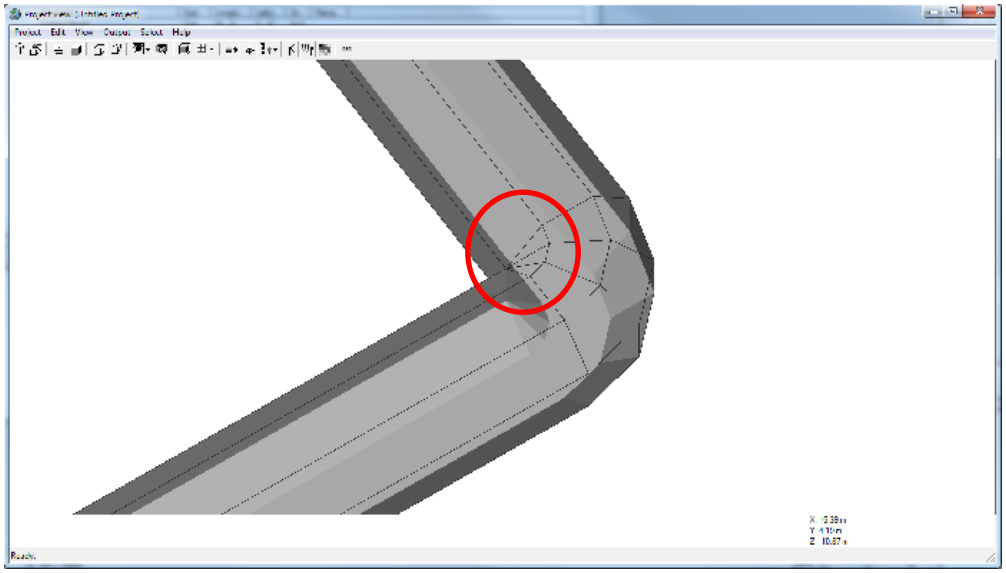


Figure 2.2.9 – *Sharp inside bend?* – Checked

2.2.5 Materials

For now, the user may only select metal or concrete for the culvert's material. To select metal, check the box labeled, *Metal? (No=concrete)*. You may change the culvert's material through the Features tab in the Wireless InSite Main window, as you would any other feature. Using concrete for the material of the culvert is suggested and has provided very good results for all bend cases.

2.2.6 Enclosing Box

The final step to create the culvert is deciding whether you would like to include an enclosing box. The enclosing box is a separate sub-structure that is introduced to prevent unwanted paths. This will allows more accurate calculations because the user can concentrate the majority of the energy in a certain area. Figure 2.2.10 displays a picture of the culvert without the Enclosing Box. Figure 2.2.11 shows the same culvert with the Enclosing Box. One known issue with X3D allows for some paths to enter the enclosed box.

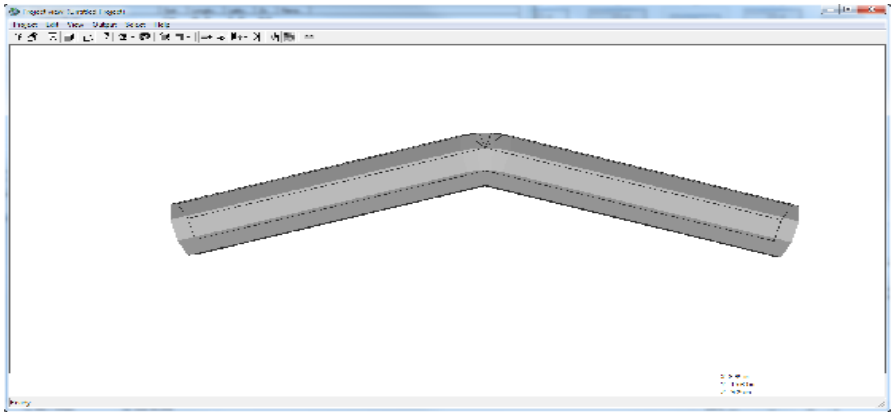


Figure 2.2.10 – Culvert without the enclosing box.

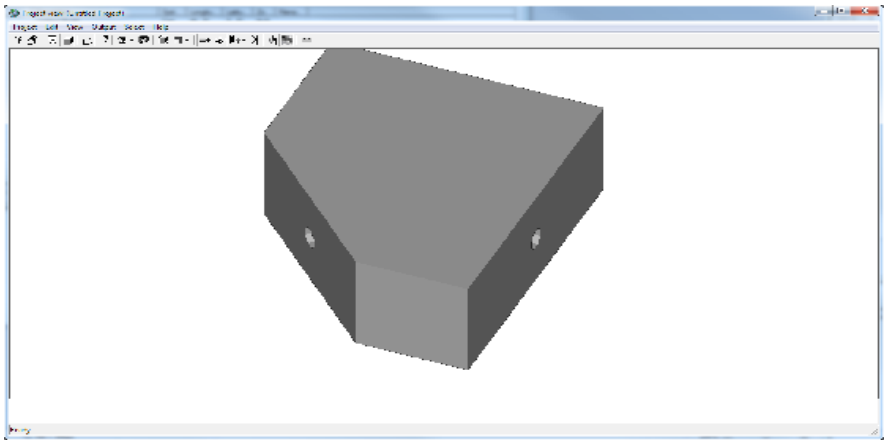


Figure 2.2.11 – Culvert with the enclosing box.

3. Study Area Set-Up

3.1 Study Area Geometry

The study area geometry can be carefully constructed to optimize the accuracy of Wireless InSite's propagation models. The propagation models (X3D and Full3D) utilize unique ray-tracing algorithms which depend on several factors. These factors often change depending on the scenario, making it impossible to choose a perfect combination of reflections and diffractions that can be used for every scenario. Furthermore, there are situations where you want to include several interactions to assure the most accurate results.

To determine the electric field inside of a culvert, it is very important to sample as many paths as possible. This suggests that the user should allow as many interactions as possible, mainly reflections. A culvert is generally small compared to other Wireless InSite Features. This introduces the possibility for a path to exit the tail end of the culvert and to re-enter at the mouth. The user may wish for such a path not to be considered. The Study Area below intersects the Enclosed Box, so that a path would need to leave the study area to make it to the other end of the culvert. Such a path will not exist using Wireless InSite's ray tracing capabilities. All paths arriving at the receiver set must go through the culvert. There may also be scenarios where the user wishes for the energy to run across the top of the culvert, or onto the opposite side. Again, with careful study area construction, this can easily be done.

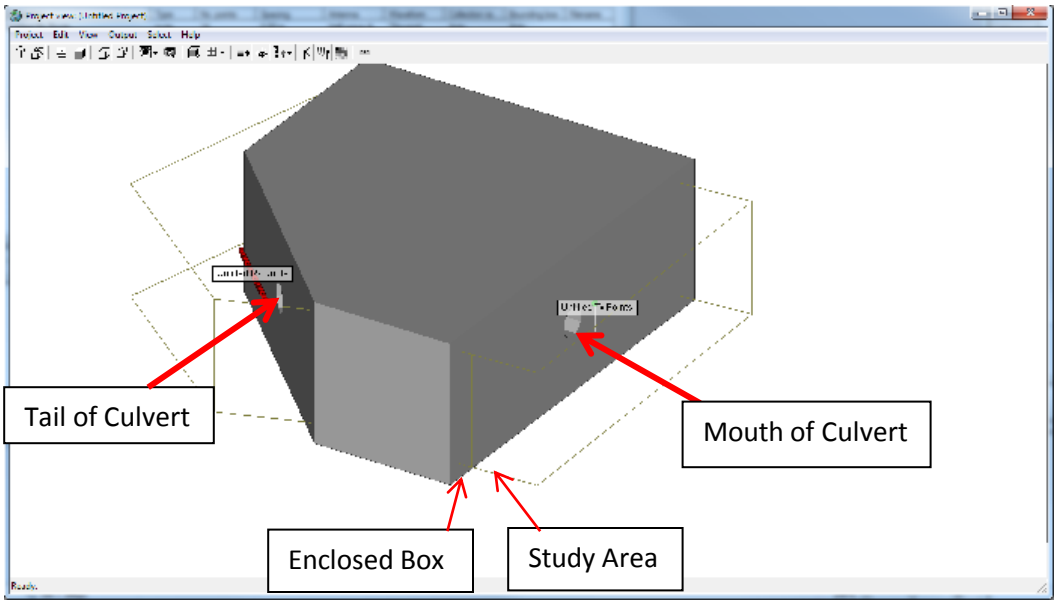


Figure 3.1 – Shows the Study Area intersecting the Enclosed Box to help eliminate paths from exiting one end of the culvert and re-entering on the opposite end.

3.2 Propagation Models; Ray Spacing and Suggested Interactions

Selecting enough interactions and sufficient ray spacing is extremely important to ensure accurate electric field calculations. As in any Wireless InSite project, the user will need to specify the amount of interactions and the density of ray spacing. This value will vary for each user because these variables will have a large effect on run-time and accuracy. To obtain results quickly, the user may want to use X3D with 10 reflections, 1 diffraction, and a route of receivers going from the beginning of the culvert, to the end. For a more accurate solution, the user should request 16-20 reflections and as many as 2 diffractions. It is usually suggested for the user to use, at least, 1 diffraction when the culvert has a bend.

The run time/accuracy trade-off is ultimately up to the user. Below is a table of suggested interactions and ray spacing which was used to validate X3D and Full3D with Remcom's XFdtd. For results displaying the accuracy of X3D and Full3D, based on different interaction and ray spacing schemes, please see ***Appendix A. Propagation Model Analysis***.

Settings	Case I-A	Case I-B	Case II-A	Case II-B
Ray Spacing (deg.)	0.05	0.05	0.05	0.05
Reflections	16	16	16	16
Transmissions	0	0	0	0
Diffractions	0	1	1	2

Table 3.2.1 – Suggested interactions and ray spacing.

Case I. Straight Culvert (no bends)

A. All Receivers (Rx) and Transmitters (Tx) are inside culvert.

B. Tx(s) or Rx(s) outside of culvert.

Case II. 90 degree bend and 45 degree bend

A. All Rx and Tx are inside of culvert.

B. Tx(s) or Rx(s) outside of culvert.

Note: This study was conducted before X3D was properly validated with XFdtd. Upon validation, it has been demonstrated that the suggested number of reflections and the ray tracing density could be decreased, resulting in faster run-times. It is important for the user to experiment using different settings to obtain a comfortable speed / accuracy trade-off.

4. Transmitter and Receiver Set-Up

Transmitters and Receivers are set up using the same techniques as any other Wireless InSite project. To obtain results within a considerable amount of time, it is suggested that the user run a receiver route along the culvert and plot the requested output. It is also possible; yet much more time consuming, to run a grid of receivers. By using a grid, the user will be able to see a visual photograph of the coverage area inside a culvert. This is most helpful situations where an operator does not have control over the placement of a receiver within a culvert.

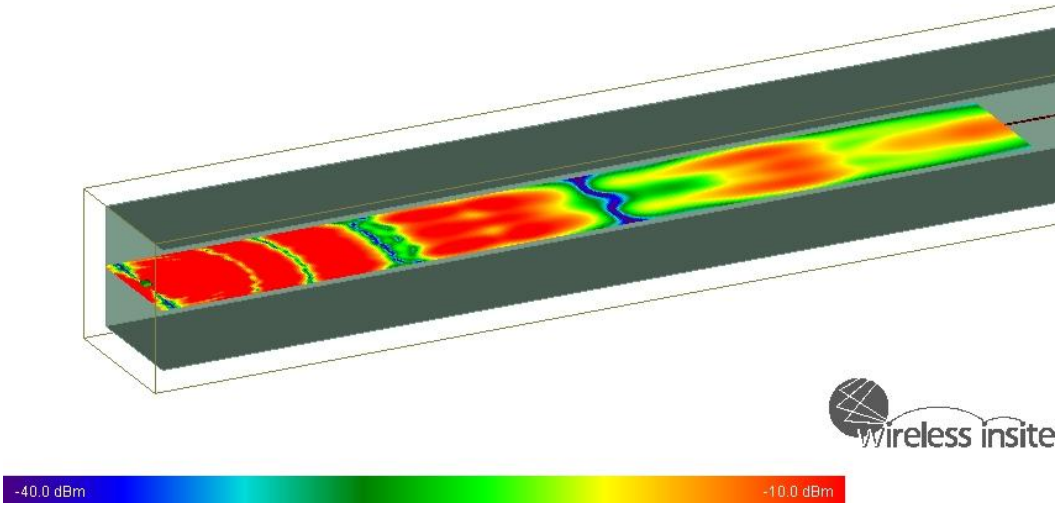


Figure 4.1 – Shows the Electric field magnitude for a grid of receivers, lying on a plane, halfway up the culvert.

Appendix

A. Propagation Model Analysis

Remcom's Wireless InSite Culvert Modeling capabilities are possible because of the ray tracing techniques provided through Remcom's propagation models X3D and Full3D. Therefore, these are the only two propagation models that should be used when the user wishes to analyze a culvert within its scenario. The user should refer to the Wireless InSite Reference Manual for specific details pertaining to the differences between X3D and Full3D. This appendix is meant to discuss the general outcome of studies conducted by Remcom engineers. Remcom's goal was to determine the validity of modeling electromagnetic waves inside a culvert when using X3D and Full3D.

A.1 Comparing X3D and Full3D with XFdtd®

To check the validity of X3D and Full3D, we chose to compare the electric field RMS to that of Remcom's XFdtd. All cases that were introduced in the *Geometry Set-Up* chapter were tested: 0, 45, and 90 degrees. It was clear that the results of Full3D compared very closely to those of XFdtd. There are a few instances where the resulting Erms plots are off by a few dB, but they usually agree very well. X3D also compares nicely, but is not quite as predictable. It often takes only a fraction of the time to simulate an X3D route of 700 receivers vs. a Full3D route of 700 receivers.

The following test cases are 45 degree vertical polarization, 45 degree horizontal polarization, and 90 degree vertical polarization. Each case considers a single transmitter close to the mouth and positioned in the very center. Each case also contains a row of receivers, positioned in the upper left hand corner of the mouth and extending from the mouth to the tail of the culvert. The culvert in each test case has a 1m x 1m cross sectional area and extends 20m. The bends are both placed at the center of the culvert.

Each plot contains XFdtd's Erms vs. Distance results, X3D's Erms vs. Distance results, and Full3D's Erms vs. Distance results. Full3D's solution was generated using 16 reflections, 1 diffraction, and 0.05 degree ray spacing. There are two plots per polarization to demonstrate how it is possible to generate favorable results with less interactions and coarser ray spacing by using X3D. The two plots for each polarization test case compares X3D with 16 reflections, 2 diffractions, and 0.25 degree ray spacing, with X3D with 10 reflections, 1 diffraction, and 0.05 degree ray spacing.

45 Degree Bend

Vertical Polarization

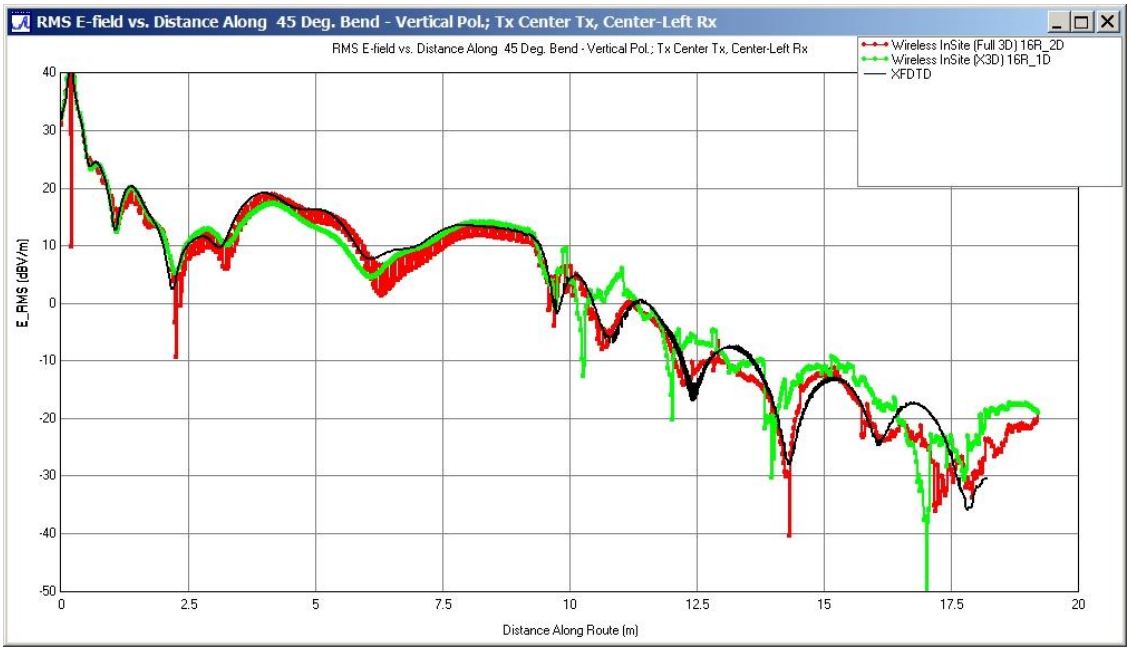


Figure A.1 – X3D: 10 Reflections, 1 Diffraction; 0.25 degree ray spacing.

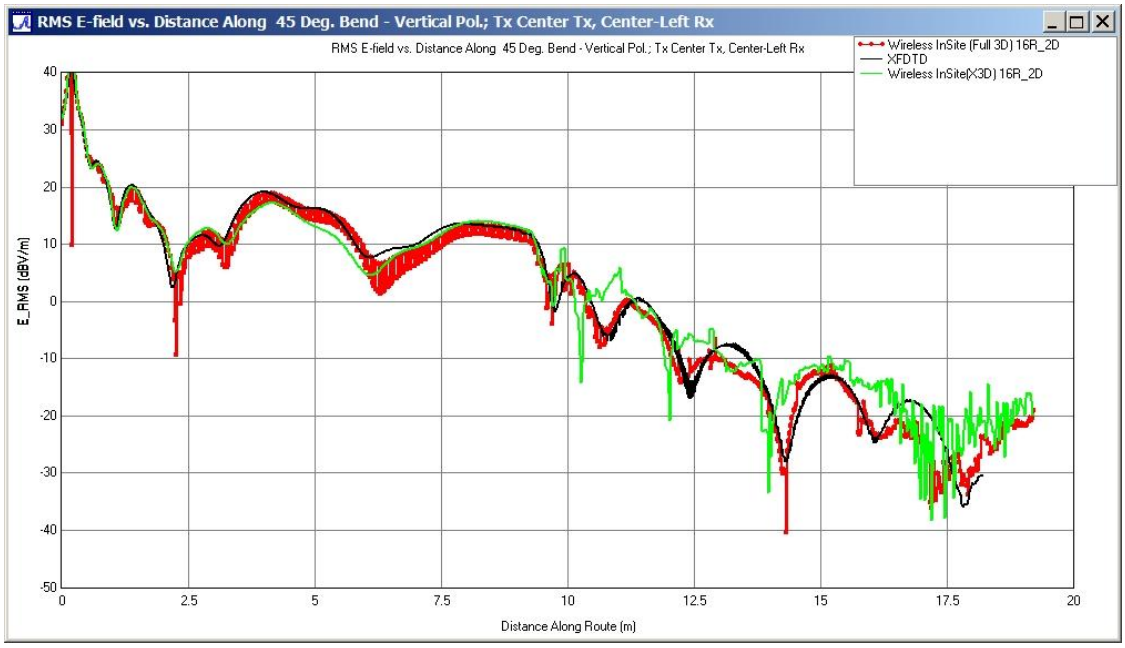


Figure A.2 – X3D: 16 Reflections, 2 Diffraction; 0.05 degree ray spacing.

Vertical Polarization

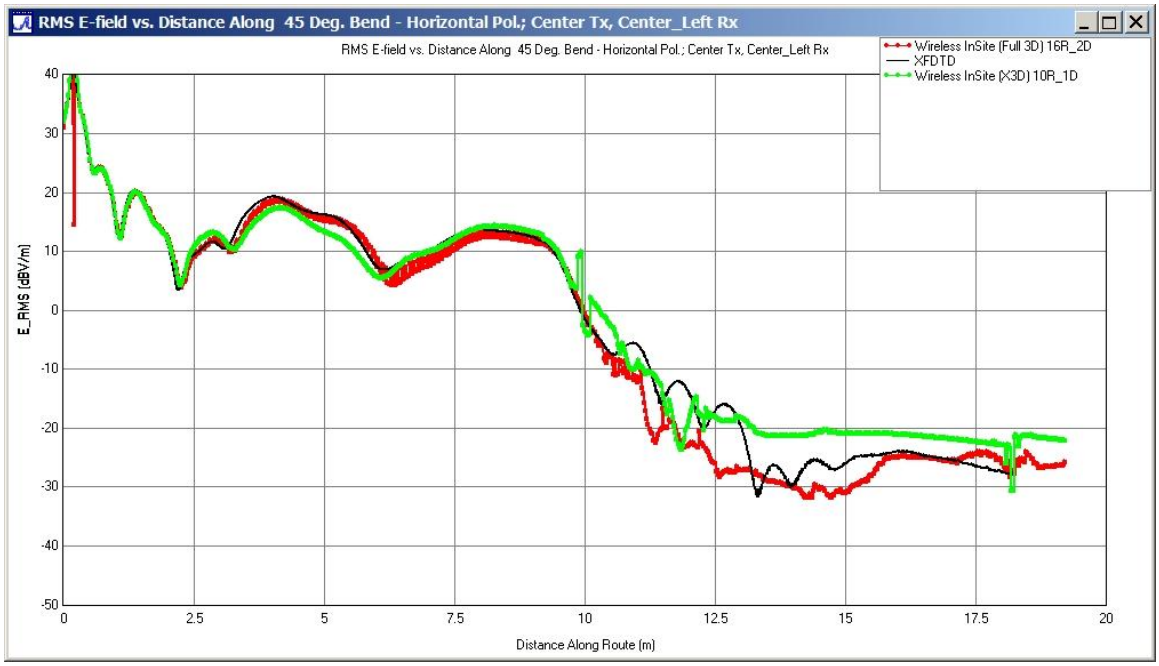


Figure A.3 – X3D: 10 Reflections, 1 Diffraction; 0.25 degree ray spacing.

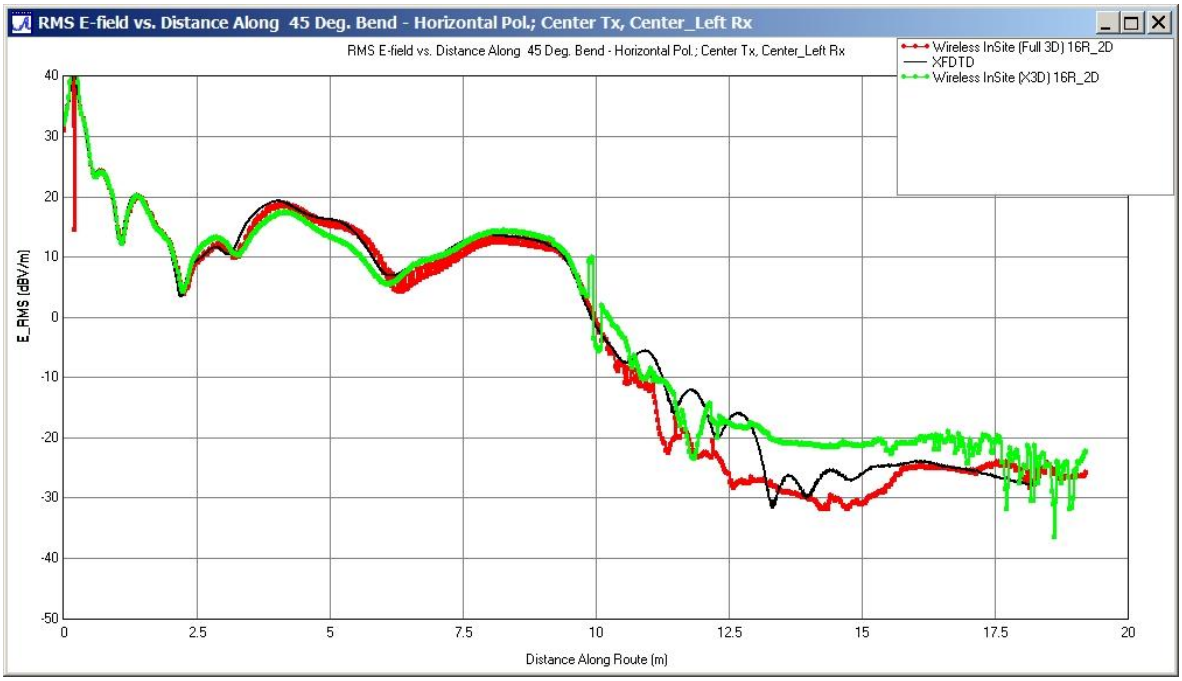


Figure A.4 – X3D: 16 Reflections, 2 Diffraction; 0.05 degree ray spacing.

90 Degree Bend

Vertical Polarization

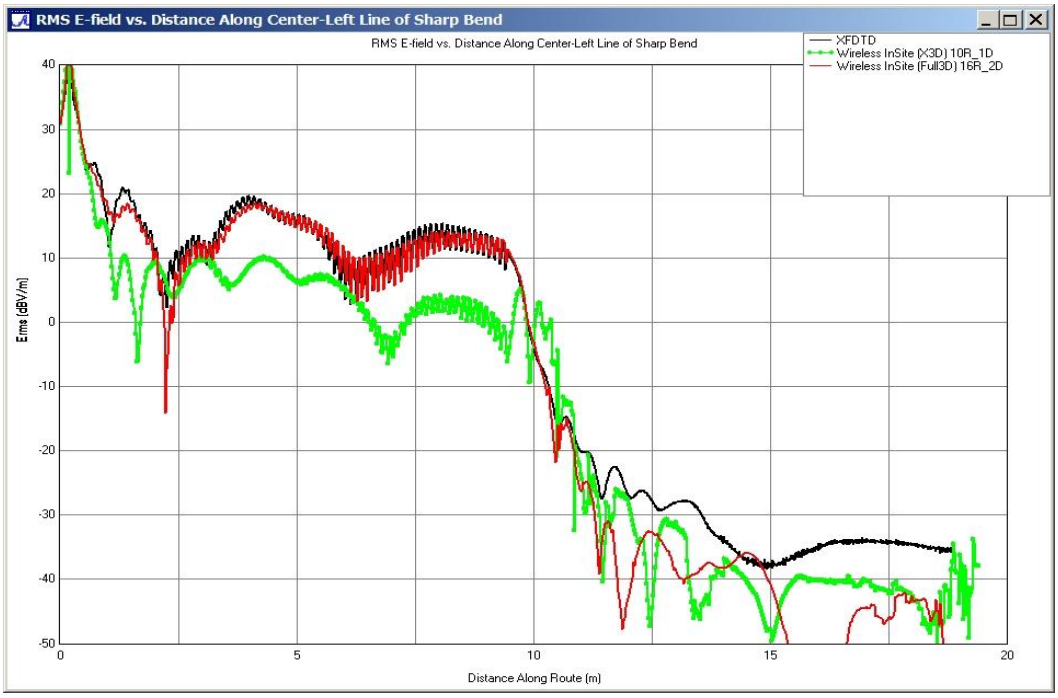


Figure A.5 – X3D: 10 Reflections, 1 Diffraction; 0.25 degree ray spacing.

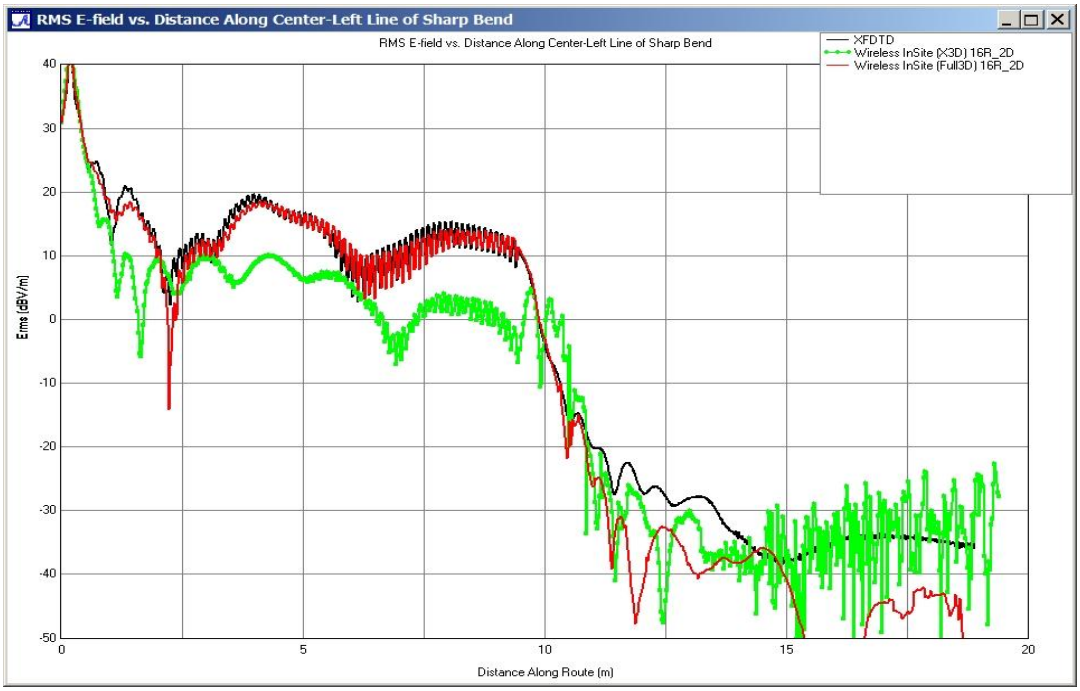


Figure A.6 – X3D: 16 Reflections, 2 Diffraction; 0.05 degree ray spacing.